

Ocean thermal energy; Geothermal energy

We are going to learn about two more important renewable energy resources.

One is Ocean Thermal Energy and another one is Geothermal energy

First, we begin with Ocean Thermal Energy

The Ocean Thermal Energy is abundantly available, and it is free

Surface ocean currents can occur as long as the sun shines.

Recent estimates suggest that ocean thermal energy itself would be sufficient to satisfy the world's electricity demand.

This is one of the non-conventional energy resources.

The ocean covers more than 70 percent of the surface of our planet.

In the oceans the temperature of water might vary by location. The latitude and the depth of the ocean or variations in solar radiation availability and also the physical properties of water determine the temperature variations of ocean water. As the solar radiation falls on the surface water Sun's heat energy will be transferred to the water. Therefore, the surface water continually becomes much warmer than the deep waters. Sunlight does not penetrate as deeply; thus, the deeper parts of the ocean are relatively cool. This causes a temperature difference between the surface water and the deep water of the sea. This temperature difference can be exploited to produce electricity. This process is known as Ocean Thermal Energy Conversion (OTEC).



Photo courtesy <https://commons.wikimedia.org/>

One of the main requirements for ocean thermal energy conversion OTEC is the temperature difference between the upper layer water and the deep sea water should be at least $>20^{\circ}\text{C}$. Mostly the temperature difference is maximum in the tropics nearly 20 to 25°C because the tropics receive a lot of sunlight throughout the year which warms the surface of the oceans hence increasing the temperature difference or temperature gradient. The efficiency of the system depends on the temperature difference. Greater the temperature difference the greater the efficiency.

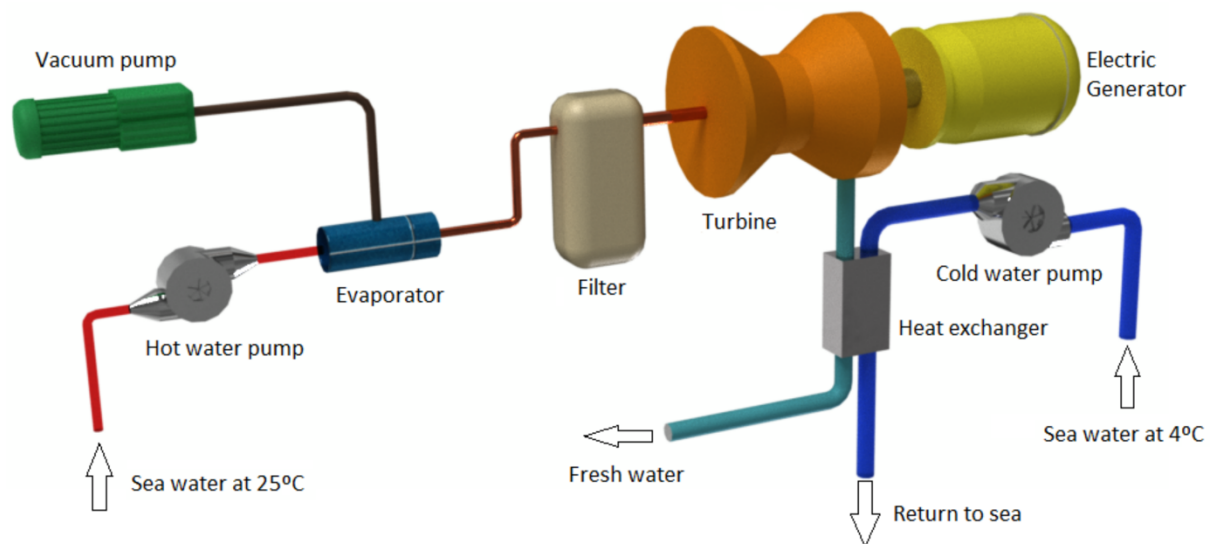
Let's discuss how to harvest the electricity from Ocean Thermal Energy Conversion (OTEC) technologies.

There are two different methodologies to harvest electricity.

The first one is closed-cycle ocean thermal energy and the second one is open-cycle thermal energy.

We can discuss about what is closed-cycle ocean thermal energy

The warm seawater would be pumped from surface warm seawater and then it will be sent to the heat exchanger. Heat exchanger is a device which has a low boiling working fluid. Mostly ammonia whose boiling point is -33°C and propane whose boiling point is -42°C or 1112-Tetrafluoroethane also called as R-134a whose boiling point is -26.3°C . Usually these fluids will be used as a refrigerant. Inside the heat exchanger heat will be absorbed from the warm water and the fluid will get vaporized. These vapors will rotate the turbine and thus the generator will generate electricity. During this process cold water from the deep sea also will be pumped to cool down the vapors. Coldwater is used to cool the vapors again into liquid. Now the fluid vapor will be converted into liquid. The same liquid is recycled back to the heat exchanger. That's why this process is known as closed-cycle ocean thermal energy conversion.



Reused from Creative Commons Attribution 4.0 licence (CC BY- 4.0), J. Mar. Sci. Eng. 2021, 9, 356 @ MDPI.

Let's discuss the second method that is open-cycle thermal energy to harvest electricity from ocean thermal energy. In this open cycle methodology, the warm seawater is first pumped into a low-pressure container. Because of the reduction in the pressure the boiling point of the water will drop. This causes the water to boil quickly and expand into vapor. This vapor or

steam passes to the turbine and rotates the low-pressure turbine which is connected directly to an electrical generator. Finally, the generator will generate electricity.

There is an advantage with this second methodology that is in the open cycle system steam will be converted back to the water during the cooling process. Finally, we are getting pure desalinated water from the plants in the form of steam. Since the steam is free of impurities this desalinated water can be used for various applications including domestic agricultural and industrial purposes.

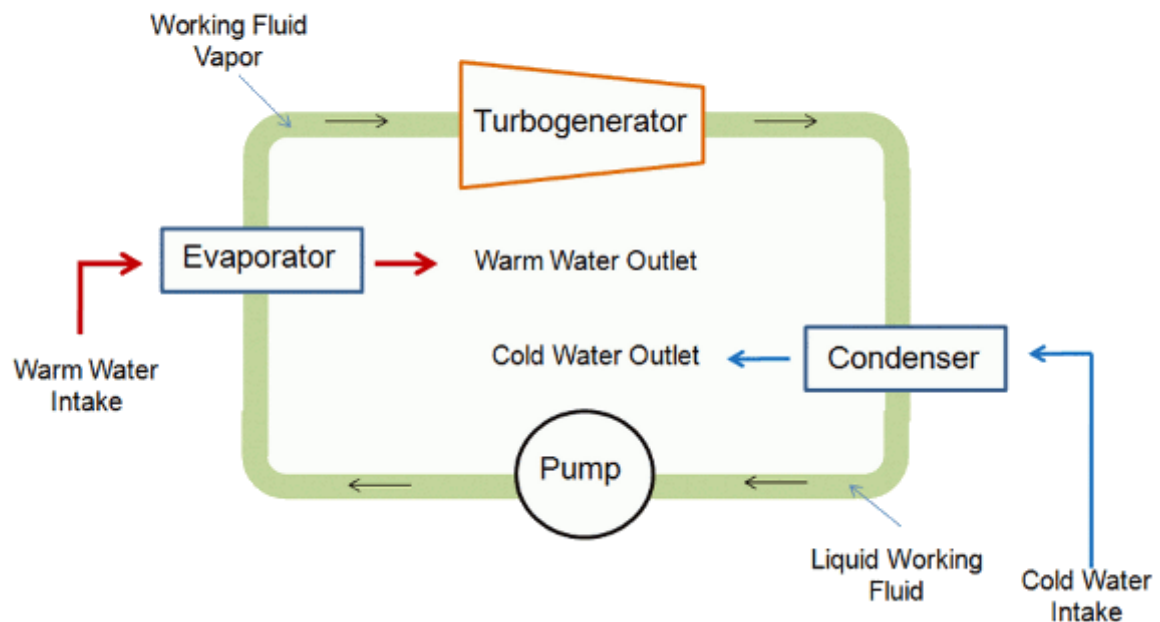


Photo courtesy to Sara E. Harrison on <http://large.stanford.edu/courses/2010/ph240/harrison2/>

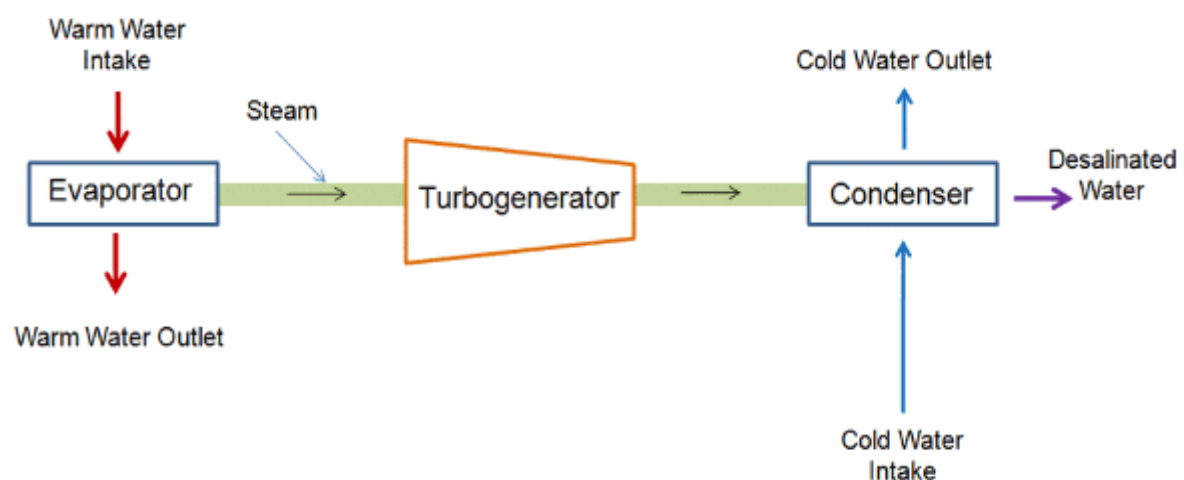


Photo courtesy to Sara E. Harrison on <http://large.stanford.edu/courses/2010/ph240/harrison2/>

Let's discuss the advantages and disadvantages of Ocean Thermal Energy.
Global demand for energy is increasing rapidly with each passing day.

We need to look for non-conventional or alternative energy resources

Ocean Thermal Energy is an excellent renewable Energy

Also, the oceans aren't going to dry up very soon.

The variance in temperatures in the oceans will almost always exist. Hence, we can generate electricity for a lifetime

Ocean Thermal Energy is renewable and clean energy. It does not cause pollution in the atmosphere or to the water bodies.

Ocean Thermal Energy is reliable. If you take wind or solar power, they are highly dependent on the weather conditions. But ocean thermal energy is reliable because as long as the sun is there the temperature variation will occur.

Ocean Thermal Energy is environmentally friendly: The ocean thermal energy plants are mainly constructed in the sea usually far away from human settlements.

Also, once the machinery and heat exchanger pipes are installed. They need no or low maintenance.

Many times, ocean thermal energy is harvested using floating-type OTEC plants that are primarily very small and hence cheaper.

Finally, the ocean thermal energy is independent of weather

Let's discuss the Disadvantages of Ocean Thermal Energy

The initial installation of the plant is an expensive process

Also pumping of either a warm seawater or deep cold water causes disruptions in aquatic and marine life.

With the available current technology, we can only achieve low efficiency nearly 1-3%.

It is location specific. Some countries lack coastal line areas and thus they cannot be able to utilize this technology.

In the case of floating type OTEC plants electricity must be transported from the ocean to land. It is expensive process.

Let's move to the second renewable energy resource that is Geothermal Energy

Geothermal energy is not an indirect form of solar energy. It is a completely new form of energy which is relied only on the heat produced under the earth mainly at the core of the earth.

Geothermal energy is the thermal energy which is stored deep inside the earth. The average increase in temperature with a depth of the earth is 1°C for every 35 to 40 meter depth.

In the earth's core the pressure and temperature are very high, and the temperature is as same as the sun which is due to the continuous nuclear fusion reactions taking place inside the earth core. Because of this some rocks melt. These molten rocks are called magma. Magma is found under the surface of the earth. Magma rises up from the earth's mantle causing upward movement. These molten rocks get trapped in particular regions which are referred to as hot spots. If the magma pushes through the cracks or holes present in the weak spots of the earth's crust it results. in a volcanic eruption.

Similarly, when the magma and underground water come in contact they form a hot spot from there steam is generated. Sometimes the steam from that region finds channels or openings at the surface. Such holes are known as hot springs or geysers. Geothermal energy is harvested mostly near hot springs or geysers or volcanic activity

Moreover, the molten magma contains water in the form of steam which is used to run turbines and hence electricity is produced.

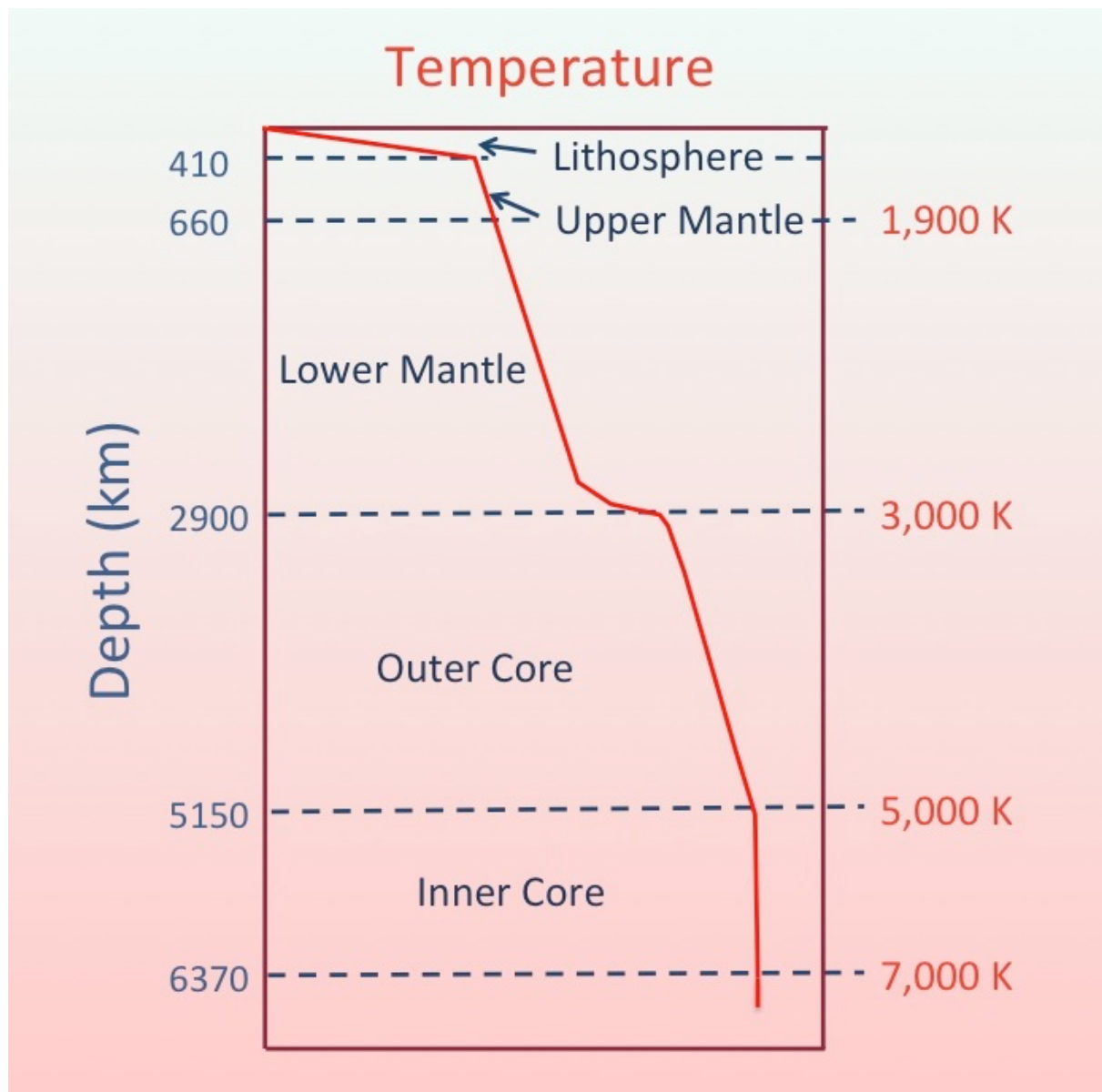


Photo courtesy <https://commons.wikimedia.org/>

Let's see how to harvest geothermal energy.

A hydrothermal convection system can be used to harness geothermal energy. That means cooler water is sent into Earth's crust where it is heated up and then rises to the surface in the form of steam.

In this process wells are drilled 1 or 2 miles deep under the Earth and the pipes are inserted. through the holes.

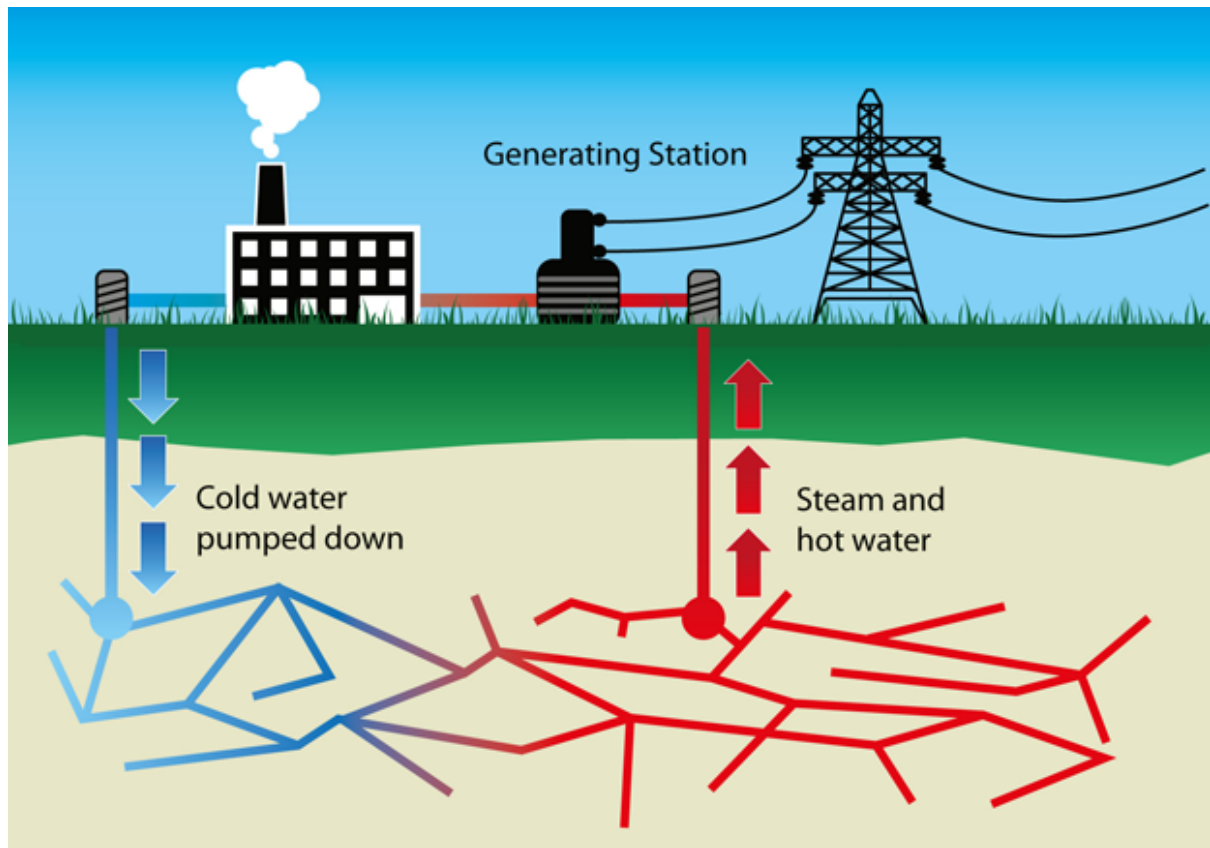


Photo courtesy <https://commons.wikimedia.org/>

Coldwater is injected deep underground through an injection well. Then it is converted into hot water. Hot water flows up through a well from deep underground due to its high pressure. When the water reaches the surface, the pressure decreases which makes the water boil subsequently converted into steam.



Photo courtesy <https://commons.wikimedia.org/>

This steam is then used to turn the blades of a turbine which is connected to a generator. This electric generator converts mechanical energy into electricity.

Geothermal power plants use cooling towers to cool-off the steam and condense it back into the water.

The cooled water is reinjected back into the Earth and the cycle begins once again.



Photo courtesy <https://commons.wikimedia.org/>

Let's discuss the advantages and disadvantages of geothermal energy

There are plenty of hot springs and natural geysers across the world that emit geothermal energy.

Geothermal energy is an inexhaustible source of energy and is available 24 hours a day 365 days a year. It is free and abundant. The earth will continuously transmit heat from its core, and it is inexhaustible and limitless. It may be available for around 2-3 billion years easily. Geothermal energy is a renewable resource: Because heat is continuously produced inside the earth due to nuclear fusion reactions. Hence the amount of heat generated by the Earth's core is unlimited.

Geothermal energy is environmentally friendly energy because it is non-polluting and self-replenishing. Also, it does not release hazardous greenhouse gases unlike fossil fuels.

Also, it does not generate any residue or any by-product.

Thus, it is safe for both environmental and human health.

Let's talk about the disadvantages of geothermal energy

Geothermal energy is an expensive one. It requires a huge initial investment for the drilling and installation of a complex systems.

Geothermal energy is location specific and limited to particular regions. Geothermal plants need to be built in geologically active areas where some geysers or hot springs or volcanic activity are available. Unlike fossil fuels geothermal energy requires large transmission lines to transport the electricity produced from it.

Geothermal sites are present deep under the earth so the process of drilling may increase earthquake risk since; earth's structure is altered during drilling or digging processes.

The drilling process results in the release of toxic gases such as sulfur dioxide and hydrogen sulfide into the environment which is harmful.